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System and method for managing data-file transmission and access right to data files

Abstract

A first user device has first public and private keys and a file encryption key for encrypting the specified file into an encrypted file. A second user device has second public and private keys. A server is in communication with the first and second user devices. The first user device generates temporary public and private keys, performs a first key establishment with the second public key, first private key and temporary private key to generate a shared secret, and uses the shared secret to encrypt the file encryption key into an encrypted key. The server allows the first user device to transmit the first public key, temporary public key, encrypted key and encrypted file to the second user device therevia if both of the first and second user devices pass the identity authentication performed thereby; and the server or the second user device uses the first public key, second private key and temporary public key to perform a second key establishment to retrieve the shared secret, and uses the shared secret, encrypted key and encrypted file to retrieve the specified file for the second user device to access.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to system and method for managing data-file transmission and access right to data files, and more particularly to system and method for managing data-file transmission and access right to data files in a communication network or a data processing system.

BACKGROUND OF THE INVENTION

(2) In current communication networks and data processing systems, it is critical to secure data files transmitted from a sender to a receiver. The data files should be protected not only from being tampered with, but also from being listened in during transmission. Among the popular technologies, the PGP (Pretty Good Privacy) encryption protocol, for example, is a widely used method of data encryption and transmission. However, it requires the user to create a password and designate an information channel to transmit the password. Therefore, it is not feasible to manage files and groups in a systemic manner. Furthermore, the file access right in a conventional file server is substantially managed by way of the user's identity. Once a design flaw or configuration oversight in the operating system or application is exploited, an advanced access privilege higher than the default access right would be taken, and unauthorized access could be executed.

SUMMARY OF THE INVENTION

(3) Therefore, the present invention addresses and overcomes the shortcomings of the vulnerability of access rights to unauthorized changes, caused by the requirements of passwords and transmission channels.

(4) In a first aspect of the present invention, a system for managing data-file transmission and access right to a specified file, comprising: a first user device having a pair of first public key and first private key and a file encryption key, which is used to encrypt the specified file into an encrypted file; a second user device having a pair of second public key and second private key; and a server in communication with the first user device and the second user device, performing identity authentication for the first user device and the second user device. For transmitting the encrypted file from the first user device to the second user device, the first user device generates a pair of temporary public key and temporary private key, acquires the second public key of the second user device, performs a first key establishment with the second public key, the first private key and the temporary private key to generate a shared secret, and uses the shared secret to encrypt the file encryption key into an encrypted key; the server allows the first user device to transmit the first public key, the temporary public key, the encrypted key and the encrypted file to the second user device therevia if both of the first and second user devices pass the identity authentication; and the server or the second user device uses the first public key, the second private key and the temporary public key to perform a second key establishment to retrieve the shared secret, and uses the shared secret, the encrypted key and the encrypted file to retrieve the specified file for the second user device to access.

(5) In a second aspect of the present invention, a method for managing data-file transmission and access right to a specified file between a first user device having a pair of first public key and first private key and a second user device having a pair of second public key and second private key via a server, comprising: the first user device using a file encryption key to encrypt the specified file into an encrypted file; the first user device generating a pair of temporary public key and temporary private key, acquiring the second public key of the second user device, performing a first key establishment with the second public key, the first private key and the temporary private key to generate a shared secret, and using the shared secret to encrypt the file encryption key into an encrypted key; the server performing identity authentication for the first user device and the second user device; and the first user device, which passes the identity authentication, transmitting the first

public key, the temporary public key, the encrypted key and the encrypted file to the second user device, which also passes the identity authentication, via the sever for the second user device to retrieve the specified file.

(6) In a third aspect of the present invention, a system for managing data-file transmission and access right to a specified file, comprising: a first user device having a pair of first public key and first private key and a file encryption key, which is used to encrypt the specified file into an encrypted file; a second user device having a pair of second public key and second private key; and a server in communication with the first user device and the second user device, performing identity authentication for the first user device and the second user device. For transmitting the encrypted file from the first user device to the second user device, the first user device acquires the second public key of the second user device, uses the second public key to encrypt the file encryption key into an encrypted key; the server allows the first user device to transmit the encrypted key and the encrypted file to the second user device therevia if both of the first and second user devices pass the identity authentication; and the server or the second user device uses the encrypted key and the encrypted file to retrieve the specified file for the second user device to access.

(7) In a fourth aspect of the present invention, a method for managing data-file transmission and access right to a specified file between a first user device having a pair of first public key and first private key and a second user device having a pair of second public key and second private key via a server, comprising: the first user device using a file encryption key to encrypt the specified file into an encrypted file; the first user device acquiring the second public key of the second user device, and using the second public key to encrypt the file encryption key into an encrypted key; the server performing identity authentication for the first user device and the second user device; and the first user device, which passes the identity authentication, transmitting the encrypted key and the encrypted file to the second user device, which also passes the identity authentication, via the sever for the second user device to retrieve the specified file.

(8) In a fifth aspect of the present invention, a system for managing data-file transmission and access right to a specified file, comprising: a first user device corresponding to a pair of first public key and first private key; a second user device corresponding to a pair of second public key and second private key; and a server in communication with the first user device and the second user device. The first user device or the server uses a file encryption key to encrypt the specified file into an encrypted file, uses the second public key to encrypt the file encryption key into a first encrypted key, processes the encrypted file into a ciphertext file having a specific file header format containing the first encrypted key, and stores the ciphertext file into the server so that the second user device is allowed to use the second private key to decrypt the ciphertext file and retrieve the specified file.

(9) In a sixth aspect of the present invention, a method for managing data-file transmission and access right to a specified file between a first user device corresponding to a pair of first public key and first private key and a second user device corresponding to a pair of second public key and second private key via a server, comprising: the first user device or the server using a file encryption key to encrypt the specified file into an encrypted file; the first user device or the server acquiring the second public key of the second user device, and using the second public key to encrypt the file encryption key into a first encrypted key; the first user device or the server processing the encrypted file into a ciphertext file having a specific file header format containing the first encrypted key, and storing the ciphertext file into the server so that the second user device is allowed to use the second private key to decrypt the ciphertext file and retrieve the specified file.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:
- (2) FIG. 1 is a scheme illustrating data encryption and transmission conducted according to an embodiment of the present invention;
- (3) FIG. 2 is a block diagram schematically illustrating a system, which conducts the data encryption and transmission illustrated in FIG. 1, according to an embodiment of the present invention;
- (4) FIG. 3 is a block diagram schematically illustrating an example of the first user device included in the system of FIG. 2, which communicates with the server via a network;
- (5) FIG. 4 is a block diagram schematically illustrating an example of the second user device included in the system of FIG. 2, which communicates with the server via a network;
- (6) FIG. 5 is a block diagram schematically illustrating an example of the sever included in the system of FIG. 2;
- (7) FIG. 6 is a scheme illustrating data encryption and transmission conducted according to another embodiment of the present invention;
- (8) FIG. 7A is a block diagram schematically illustrating a system, which conducts the data encryption and transmission illustrated in FIG. 6, according to an embodiment of the present invention;
- (9) FIG. 7B is a block diagram schematically illustrating an example of the ciphertext file included in the server of FIG. 7A;
- (10) FIG. 8A is a block diagram schematically illustrating a system, which conducts the data encryption and transmission illustrated in FIG. 6, according to another embodiment of the present invention; and
- (11) FIG. 8B is a block diagram schematically illustrating an example of the ciphertext file included in the server of FIG. 8A.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

- (12) The invention will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this invention are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.
- (13) Data encryption and transmission executed by a system for managing data-file and access right to data files according to an embodiment of the present invention is described hereinafter with reference to the scheme of FIG. 1. In Steps **111**, **112** and **113**, which can be executed in an arbitrary order, the sender **11** prepares a file F to be transmitted, acquires a public key PU.sub.b of a receiver **12**, and generates a file encryption key d.sub.k. Then in Step **114**, the file F is encrypted with the file encryption key d.sub.k to generate an encrypted file [E(d.sub.k, F)]. The encryption in Step **114** can be executed by Authenticated Encryption with Associated Data (AEAD). For example, the currently available Advanced Encryption Standard-Galois/Counter Mode (AES-GCM) can be used for encryption, and meanwhile assure of the security, integrity and identifiability of the file F. Subsequently in Step **115**, key establishment is performed to obtain a shared secret SS. The key establishment may be conducted based on a key agreement protocol, e.g., Elliptic Curve Diffie-Hellman key exchange(2s,1e), which is also known as ECDH(2s,1e)). For example, the shared secret SS of the sender **11** is established with the public key PU.sub.b of the receiver **12** mentioned in Step **112**, the private key PR.sub.a of the sender **11** and the temporary private key PR.sub.e of the sender **11**. In Step **116**, the encrypted key [E(SS, d.sub.k)] is formed by encrypting the file encryption key d.sub.k with the shared secret established in Step **115**. Afterwards, the sender **11** processes a temporary public key PU.sub.e of the sender **11**, the encrypted file [E(d.sub.k, F)] and the encrypted key [E(SS, d.sub.k)] into a data packet (PU.sub.e, [E(d.sub.k, F)], [E(SS, d.sub.k)])

and transmits the data packet to the receiver **12** (Step **117**). The pair of temporary public key PU.sub.e and temporary private key PR.sub.e of the sender **11** can only be consecutively used a limited number of times, or even once, and then discarded. A new pair of temporary public and private keys will be generated when the sender **11** is going to send another encrypted file.

(14) Meanwhile, the receiver **12**, which receives the data packet (PU.sub.e, [E(d.sub.k, F)], [E(SS, d.sub.k)]), executes a request to obtain a public key PU.sub.a of the sender **11** (Step **121**) for key establishment (Step **122**). Then the receiver **12** performs another key establishment with the public key PU.sub.a of the sender **11** mentioned in Step **121**, a private key PR.sub.b of the receiver **12**, and the temporary public key PU.sub.e received from the sender **11** to retrieve the shared secret SS. With the shared secret SS, the receiver **12** can decrypt the encrypted key [E(SS, d.sub.k)] to retrieve the file encryption key d.sub.k (Step **123**). The file encryption key d.sub.k is then used to decrypt the encrypted file so as to retrieve the file F (Step **124**).

(15) Please refer to FIG. 2, which schematically illustrates an embodiment of data file transmission system adapted to execute the data encryption and transmission illustrated in FIG. 1 according to the present invention. The system includes a first user device **21**, a second user device **22** and a server **23**. The server **23** communicates with the first user device **21** and the second user device **22** via a network, e.g., Internet. The first user device **21** generates a first pair of public and private keys, and the second user device **22** generates a second pair of public and private keys. In the following example for illustrating the principle and the practical operation of the present invention, the first user device **21** serves as the sender **11**, and the second user device **22** serves as the receiver **12**. That is, the first user device **21** generates the first public key PU.sub.a and the first private key PR.sub.a, and the second user device **22** generates the second public key PU.sub.b and the second private key PR.sub.b. For securely transmitting the file F to the second user device **22**, the first user device **21** acquires the public key PU.sub.b from the second user device **22** directly or via the server **23**, and encrypts the file F with a file encryption key d.sub.k to generate an encrypted file [E(d.sub.k, F)]. The first user device **21** uses the second public key PU.sub.b, the first private key PR.sub.a and the temporary private key PR.sub.e to establish the shared secret SS, which is used for encrypting the file encryption key d.sub.k into an encrypted key [E(SS, d.sub.k)], according to the key-agreement protocol ECDH(2s, 1e). Once the first user device **21** and the second user device **22** both pass the authentication required by the server **23**, the server **23** allows the first user device **21** to send the encrypted file [E(d.sub.k, F)], as well as the first public key PU.sub.a, the temporary private key PR.sub.e and the encrypted key [E(SS, d.sub.k)], to the second user device **22** via the server **23**. The second user device **22** uses its own second private key PR.sub.b and the first public key PU.sub.a and temporary private key PR.sub.e received from the first user device **21** via the server **23** to conduct key establishment, thereby retrieving the shared secret SS. The second user device **22** then uses the shared secret SS to retrieve the file encryption key d.sub.k from the encrypted key [E(SS, d.sub.k)], and uses the file encryption key d.sub.k to retrieve the file F from the encrypted file [E(d.sub.k, F)]. In an alternative example, after the server **23** receives the encrypted file [E(d.sub.k, F)], the first public key PU.sub.a, the temporary private key PR.sub.e and the encrypted key [E(SS, d.sub.k)] from the first user device **21**, the server **23** conducts key establishment with the first public key PU.sub.a and temporary private key PR.sub.e received from the first user device **21** and the second private key PR.sub.b provided by the second user device **22** to retrieve the shared secret SS, and uses the shared secret SS, the encrypted key [E(SS, d.sub.k)] and the encrypted file [E(d.sub.k, F)] to retrieve the file F as mentioned above. Afterwards, the server **23** transfers the file F to the second user device **22** to complete the secured file transmission.

(16) An embodiment of the first user device **21** is now illustrated with reference to the block diagram shown in FIG. 3. The first user device **21** includes a host **210** and a detachable identity authentication and encryption device **211**. The host **210** communicates with the server **23** via a network **30** such as Internet or a wide area network. The identity authentication and encryption device **211** communicates with the host **210** via a bus interface **212** when the host **210** and the

identity authentication and encryption device **211** are both electrically connected to the bus interface **212**. The bus interface **212**, for example, may be a USB interface or any other suitable bus interface. The identity authentication and encryption device **211** stores therein the first private key PR.sub.a, e.g., stored in a secure element **2110** thereof, and releases the first public key PU.sub.a pairing the first private key PR.sub.a. The identity authentication and encryption device **211** can use the first private key PR.sub.a to pass an authentication procedure. For example, the authentication procedure can be executed by the server **23**, which uses a previously obtained identity public key PU.sub.1 of a specified device to verify a digital signature derived from an identity private key PR.sub.1 of the specified device. Details of the authentication procedure can be further realized from a co-pending U.S. patent application Ser. No. 17/577,411, which is incorporated herein for reference. After the authentication procedure, the secure element **2110** of the identity authentication and encryption device **211** establishes the shared secret SS with the second public PU.sub.b, which is released by the second user device **22**, and the first private PR.sub.a and temporary private key PR.sub.e of the first user device **21** according to the key-agreement protocol ECDH(2s, 1e). Afterwards, the shared secret SS is used to encrypt the file encryption key d.sub.k to generate the encrypted file [E(d.sub.k, F)], which is then transmitted to the server **23** or further transferred to the second user device **22** via the server **23**.

(17) Furthermore, after the server **23** determines that the identity authentication and encryption device **211** passes the authentication procedure, e.g., by way of verifying the digital signature as mentioned above, the host **210** runs a browser program **32** to have the display **31** show a user interface **33**, which is originated from the server **23**. Via the user interface **33**, a user may input commands, which are differentially authorized by the server **23** to different users, to access and/or transfer files. For achieving a high level of information security, the WebAuthn standard of the commercially available FIDO2 (Fast Identity Online 2) protocol can be used as an example of the communication protocol. Accordingly, the above-described authentication process should conform to the FIDO2 protocol.

(18) In an embodiment, the second user device **22**, as a receiver, may have functional means similar to those of the first user device **21** as described above. In another embodiment, the second user device **22** does not have the identity authentication and encryption device **211**, and instead, the second user device **22** may be implemented with the one illustrated in FIG. 4. As shown, the second user device **22** includes a host **220** and a detachable identity authentication and encryption device **221**. The host **220** communicates with the server **23** via the network **30**, e.g., Internet or wide area network. The identity authentication and encryption device **221** communicates with the host **220** via a bus interface **222** when the host **220** and the identity authentication and encryption device **221** are both electrically connected to the bus interface **222**. The bus interface **222**, for example, may be a USB interface or any other suitable bus interface. When the identity authentication and encryption device **221** passes the authentication conducted by the server **23**, the host **210** runs a browser program **42** to have a display **41** show a user interface **33**, which is originated from the server **23**. Via a user interface **43**, a user may input commands, which are differentially authorized by the server **23** to different users, to access and/or transfer files. Likewise, the WebAuthn standard of the commercially available FIDO2 (Fast Identity Online 2) protocol can be used as an example of the communication protocol. Accordingly, the above-described authentication process should conform to the FIDO2 protocol. The hosts **210** and **220** used in the above embodiments may be personal computers, tablet computers, smart phones, or any other similar information-processing hosts.

(19) In this embodiment, the identity authentication and encryption device **211** does not have encryption/decryption operating means, so the decryption operations at the second user device **22** are executed by a cipher server **230** included in the server **23**. The cipher server **230** uses the first public PU.sub.a released by the first user device **21**, the second private key PR.sub.b of the second user device **22** and the temporary private key PU.sub.e of the first user device **21** to establish the

shared key SS. The cipher server **230** then decrypts the encrypted key $[E(SS, d.sub.k)]$ with the shared secret SS, retrieves the file encryption key $d.sub.k$ with the encrypted key $[E(SS, d.sub.k)]$, and retrieves the file F with the file encryption key $d.sub.k$. The file F is then stored into a file server **231** included in the server **23** or transferred to the second user device **22**.

(20) Please refer to FIG. 5, which schematically illustrates a block diagram of an embodiment of the server **23**. The server **23** includes a cipher server **230**, a file server **231** and a FIDO server **232**. The cipher server **230** functions for performing the encryption/decryption operation for a detachable identity authentication and encryption device that does not perform any encryption/decryption operation. The file server **231** functions for temporarily storing the encrypted file and the user interface program to run in the browser at the user end. The file server **231** further works as a relying party application server (RP app server), which functions for registration and verification of users. The FIDO server **232** can be disposed at a distant end, and shared with another server in addition to the server **23**.

(21) FIG. 6 shows another embodiment of data encryption and transmission conducted according to the present invention. In Steps **611**, **612** and **613**, which can be executed in an arbitrary order, the sender **11** prepares a file F to be transmitted, acquires a public key $PU.sub.b$ of a receiver **12**, and generates a file encryption key $d.sub.k$. Then in Step **614**, the file F is encrypted with the file encryption key $d.sub.k$ to generate an encrypted file $[E(d.sub.k, F)]$. The encryption in Step **614** can be executed by Authenticated Encryption with Associated Data (AEAD). For example, the currently available Advanced Encryption Standard-Galois/Counter Mode (AES-GCM) can be used for encryption, and meanwhile assure of the security, integrity and identifiability of the file F.

(22) Subsequently in Step **615**, key transport is performed. The sender **11** uses the public key $PU.sub.b$ of the receiver **12** to encrypt the file encryption key $d.sub.k$ into an encrypted key $[E(PU.sub.b, d.sub.k)]$. The sender **11** further uses a private key $PR.sub.a$ or identity private key $PR.sub.1$ thereof to make a digital signature [Sig] for the encrypted file $[E(d.sub.k, F)]$ (Step **616**). Afterwards, the sender **11** incorporates the digital signature [Sig], the encrypted file $[E(d.sub.k, F)]$ and the encrypted key $[E(PU.sub.b, d.sub.k)]$ into a data packet ($[Sig], [E(d.sub.k, F)], [E(PU.sub.b, d.sub.k)]$), and transmits the data packet ($[Sig], [E(d.sub.k, F)], [E(PU.sub.b, d.sub.k)]$) to the receiver **12** (Step **617**).

(23) On the other hand, the receiver **12** acquires a public key $PU.sub.a$ or an identity public key $PU.sub.1$ of the sender **11** in Step **621**, and executes an verifying procedure of the digital signature [Sig] in Step **622**. When it is determined, by way of the public key $PU.sub.a$ or the identity public key $PU.sub.1$ of the sender **11** and the encrypted file $[E(d.sub.k, F)]$, that the digital signature [Sig] is correct, the receiver **12** retrieves the file encryption key $d.sub.k$ from the encrypted key $[E(PU.sub.b, d.sub.k)]$ with a private key $PR.sub.b$ of the receiver **12** in Step **623**. Subsequently in Step **624**, the receiver **12** retrieves the file F from the encrypted file $[E(d.sub.k, F)]$ with the file encryption key $d.sub.k$. In this embodiment, key transport, instead of key agreement, is used to provide the key for the receiver to retrieve the file.

(24) In the case of more than one receiver, the same key combination, which consists of file encryption key $d.sub.k$, temporary public key $PU.sub.e$ of the sender and temporary private key $PR.sub.e$ of the sender, can be used to support transmission to the multiple receivers. For example, by way of key agreement, the same file encryption key $d.sub.k$ and encrypted file $[E(d.sub.k, F)]$ to be transmitted from a sender A to multiple receivers, e.g., B, C and D, can be processed with the temporary private key $PR.sub.e$ of the sender in respective key agreements between the sender A and the receivers B, C and D to generate respective shared secrets $SSab$, $SSac$ and $SSad$. In this embodiment, after respective encrypted keys $[E(SSab, d.sub.k)]$, $[E(SSac, d.sub.k)]$ and $[E(SSad, d.sub.k)]$ are formed by encrypting the file encryption key $d.sub.k$ with the shared secrets, the sender A processes the temporary public key $PU.sub.e$ of the sender A, the encrypted file $[E(d.sub.k, F)]$ and the encrypted keys $[E(SSab, d.sub.k)]$, $[E(SSac, d.sub.k)]$ and $[E(SSad, d.sub.k)]$ into a data packet, and transmits the data packet to each of the receivers. Accordingly, it is

simple to perform the secured key and file transmission to one more receiver by conducting an additional key agreement between the sender and the additional receiver to generate a shared secret between the sender and the additional receiver. The shared secret is used to form an encrypted key specific to the additional receiver. Then the encrypted key specific to the additional receiver is added to the data packet containing the encrypted keys of the other receivers.

(25) In the above embodiments, the sender is a single user device, i.e., the first user device **21**, and the receiver is also a single user device, i.e., the second user device **22**. Alternatively, the sender may also be a first group of user devices having a common pair of first public key and first private key, and the receiver may be a second group of user devices having a common pair of second public key and second private key. Since multiple user devices correspond to the same pair of public and private keys, the resource and cost for establishing multiple pairs of public and private keys can be saved. Details will be described as follows.

(26) Please refer to FIG. 7A, in which a system for managing data-file transmission and access right to data files according to another embodiment of the present invention is illustrated. The system in this embodiment includes a first group of user devices **71** having a common pair of first public key and first private key, a second group of user devices **72** having a common pair of second public key and second private key, and a server **73**. The first public key and the second public key can be uploaded for exchange. The server **73** is in communication with the first group of user devices **71** and the second group of user devices **72**. A ciphertext file **731**, which contains an encrypted file $[E(d.sub.k, F)]$ and has a specific file header format **7311**, can be stored in the server **73**, wherein the encrypted file $[E(d.sub.k, F)]$ is generated by encrypting a file F to be transmitted with a file encryption key $d.sub.k$.

(27) Furthermore, the specific file header format **7311** at least includes an encrypted key $[E(PU, d.sub.k)]$, which is generated by the server **73** or a user device by encrypting the encryption key $d.sub.k$ with a public key PU , as illustrated in FIG. 7B. Therefore, a user device, which owns a private key paired with the public key PU , would have the right to open the ciphertext file **731**. For example, if the encrypted key $[E(PU, d.sub.k)]$ of the file header format **7311** of the ciphertext file **731** is generated by encrypting the file encryption key $d.sub.k$ with a first public key, then any user device in the first group can use the common first private key paired with the common first public key to open the ciphertext file **731**. In other words, the right of each user device to access a specified file can be effectively managed by encrypting the file with a proper public key. The encryption of the file may also be executed by the server or the user device. In addition, the risk that an advanced access right is improperly acquired due to design flaws or configuration oversights existing in the operational system or application programs can be avoided, thereby eliminating security vulnerability as executing unauthorized access actions.

(28) Please refer to FIG. 8A, in which a system for managing data-file transmission and access right to data files according to a further embodiment of the present invention is illustrated. The system in this embodiment includes a first user devices **801** having a pair of first public key and first private key, a second user devices **802** having a pair of second public key and second private key, and a third user devices **803** having a pair of third public key and third private key, which are in communication with the server **73**. An encrypted file **81** is generated by the first user device **801** or the server **73** by encrypting a file F to be transmitted with a file decryption key by the first user device **801** or the server **804**. Meanwhile, the file decryption key is encrypted with the second public key and the third public key into an encrypted key **82** and an encrypted key **83**, respectively. Then a ciphertext file **732** is derived from the encrypted file **81** by way of the encrypted key **82** and the encrypted key **83**, and stored into the server **73**. Afterwards, the second user device **802** or the third user device **803** can decrypt the ciphertext file **732** with its own second private key or third private key to retrieve the file F . It is to be noted that in this embodiment, the ciphertext file **732** has a default file header format **7321**, which includes the encrypted key **82** and the encrypted key **83** at the same time. Of course, the default file header format **7321** may include more encrypted keys in

addition to the encrypted key **82** and the encrypted key **83** when more user devices are involved for transmission of the file **F**.

(29) The default file header format **7321** mentioned above may be of a constant bit-length no matter how many encrypted keys are included. Such a design is advantageous in hiding the count and forms of keys from being reckoned by non-related persons, thereby further enhancing information security. Furthermore, like the embodiments illustrated in FIGS. **3**, **4** and **5**, an external identity authentication and encryption device can be detachably connected to any of the user devices via a bus interface and used to conduct identity authentication. The identity authentication and encryption device stores therein the private key of the corresponding user device. Once the server **73** determines that the identity authentication and encryption device passes the authentication, e.g., by way of verifying the digital signature as mentioned above, the corresponding user device automatically runs a browser program to have the display show a user interface, which is originated from the server **73**. Via the user interface, a user may input data or commands.

(30) While the invention has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A system for managing data-file transmission and access right to a specified file, comprising: a first user device having a pair of first public key and first private key and a file encryption key, which is used to encrypt the specified file into an encrypted file; a second user device having a pair of second public key and second private key; and a server in communication with the first user device and the second user device, performing identity authentication for the first user device and the second user device, wherein for transmitting the encrypted file from the first user device to the second user device, the first user device generates a pair of temporary public key and temporary private key, acquires the second public key of the second user device, performs a first key establishment with the second public key, the first private key and the temporary private key to generate a shared secret, and uses the shared secret to encrypt the file encryption key into an encrypted key; the server allows the first user device to transmit the first public key, the temporary public key, the encrypted key and the encrypted file to the second user device therevia if both of the first and second user devices pass the identity authentication; and the server or the second user device uses the first public key, the second private key and the temporary public key to perform a second key establishment to retrieve the shared secret, and uses the shared secret, the encrypted key and the encrypted file to retrieve the specified file for the second user device to access.
2. The system according to claim 1, wherein the first user device includes a host and a detachable identity authentication and encryption device in communication with the host via a bus interface, wherein the identity authentication and encryption device stores therein the first private key, which is used to pass the identity authentication, generates the file encryption key and the pair of temporary public key and temporary private key, uses the file encryption key to encrypt the specified file into the encrypted file, acquires the second public key, performs the first key establishment with the second public key, the first private key and the temporary private key to generate the shared secret, uses the shared secret to encrypt the file encryption key into the encrypted key, and transmits the first public key, the temporary public key, the encrypted key and the encrypted file to the second user device, which also passes the identity authentication, via the server.

3. The system according to claim 1, wherein the second user device includes a detachable identity authentication and encryption device; when the identity authentication and encryption device is detachably connected to a bus interface of the second user device, the server executes the identity authentication of the second user device; and the second user device automatically runs a browser program to have a display show a user interface for a user to input data if the second user device passes the identity authentication.
4. The system according to claim 3, wherein the server is a server set including a cipher server, a file server and a Fast Identity Online (FIDO) server, wherein the cipher server performs encryption/decryption for the detachable identity authentication and encryption device, the file server temporarily stores the encrypted file and the browser program, and works as a relying party application server (RP app server) for registration and verification of users, and the FIDO server is disposed at a distant end and allowed to be shared with an external server.
5. A system for managing data-file transmission and access right to a specified file, comprising: a first user device having a pair of first public key and first private key and a file encryption key, which is used to encrypt the specified file into an encrypted file; a second user device having a pair of second public key and second private key; and a server in communication with the first user device and the second user device, performing identity authentication for the first user device and the second user device, wherein for transmitting the encrypted file from the first user device to the second user device, the first user device acquires the second public key of the second user device, uses the second public key to encrypt the file encryption key into an encrypted key; the server allows the first user device to transmit the encrypted key and the encrypted file to the second user device therevia if both of the first and second user devices pass the identity authentication; and the server or the second user device uses the encrypted key and the encrypted file to retrieve the specified file for the second user device to access.
6. The system according to claim 5, wherein the first user device further uses the first private key or an identity private key to make a digital signature for the encrypted file, and the digital signature is transmitted to the second user device via the server along with the first public key or an identity public key pairing the identity private key, the encrypted key and the encrypted file; and the server or the second user device uses the first public key or identity public key and the encrypted file to verify the digital signature.
7. The system according to claim 6, wherein the first user device includes a host and a detachable identity authentication and encryption device in communication with the host via a bus interface, wherein the identity authentication and encryption device stores therein the first private key, which is used to pass the identity authentication, generates the file encryption key, uses the file encryption key to encrypt the specified file into the encrypted file, uses the first private key or the identity private key to make the digital signature for the encrypted file, uses the second public key to encrypt the file encryption key into the encrypted key, and transmits the digital signature to the second user device, which also passes the identity authentication, via the server along with the first public key or identity public key, the encrypted key and the encrypted file.
8. The system according to claim 5, wherein the second user device includes a detachable identity authentication and encryption device; when the identity authentication and encryption device is detachably connected to a bus interface of the second user device, the server executes the identity authentication of the second user device; and the second user device automatically runs a browser program to have a display show a user interface for a user to input data if the second user device passes the identity authentication.
9. The system according to claim 8, wherein the server is a server set including a cipher server, a file server and a Fast Identity Online (FIDO) server, wherein the cipher server performs encryption/decryption for the detachable identity authentication and encryption device, the file server temporarily stores the encrypted file and the browser program, and works as a relying party application server (RP app server) for registration and verification of users, and the FIDO server is

disposed at a distant end and allowed to be shared with an external server.

10. A system for managing data-file transmission and access right to a specified file, comprising: a first user device corresponding to a pair of first public key and first private key; a second user device corresponding to a pair of second public key and second private key; and a server in communication with the first user device and the second user device, wherein the first user device or the server uses a file encryption key to encrypt the specified file into an encrypted file, uses the second public key to encrypt the file encryption key into a first encrypted key, processes the encrypted file into a ciphertext file having a specific file header format containing the first encrypted key, and stores the ciphertext file into the server so that the second user device is allowed to use the second private key to decrypt the ciphertext file and retrieve the specified file.

11. The system according to claim 10, further comprising a third user device being in communication with the server and sharing the pair of second public key and second private key with the second user device so that the third user device is allowed to use the second private key to decrypt the ciphertext file and retrieve the specified file.

12. The system according to claim 10, further comprising a third user device being in communication with the server and corresponding to a pair of third public key and third private key, wherein the first user device or the server further uses the third public key to encrypt the file encryption key into a second encrypted key, and the specific file header format of the ciphertext file contains the second encrypted key in addition to the first encrypted key, so that the third user device is allowed to use the third private key to decrypt the ciphertext file and retrieve the specified file.

13. The system according to claim 12, wherein the specific file header format of the ciphertext file has a constant bit length, which does not change with addition of the second encrypted key.

14. The system according to claim 10, wherein the first user device includes a host and a detachable identity authentication and encryption device in communication with the host via a bus interface, and the identity authentication and encryption device stores therein the first private key, which is used to pass the identity authentication.

15. The system according to claim 10, wherein the second user device includes a detachable identity authentication and encryption device; when the identity authentication and encryption device is detachably connected to a bus interface of the second user device, the server executes the identity authentication of the second user device; and the second user device automatically runs a browser program to have a display show a user interface for a user to input data if the second user device passes the identity authentication.
